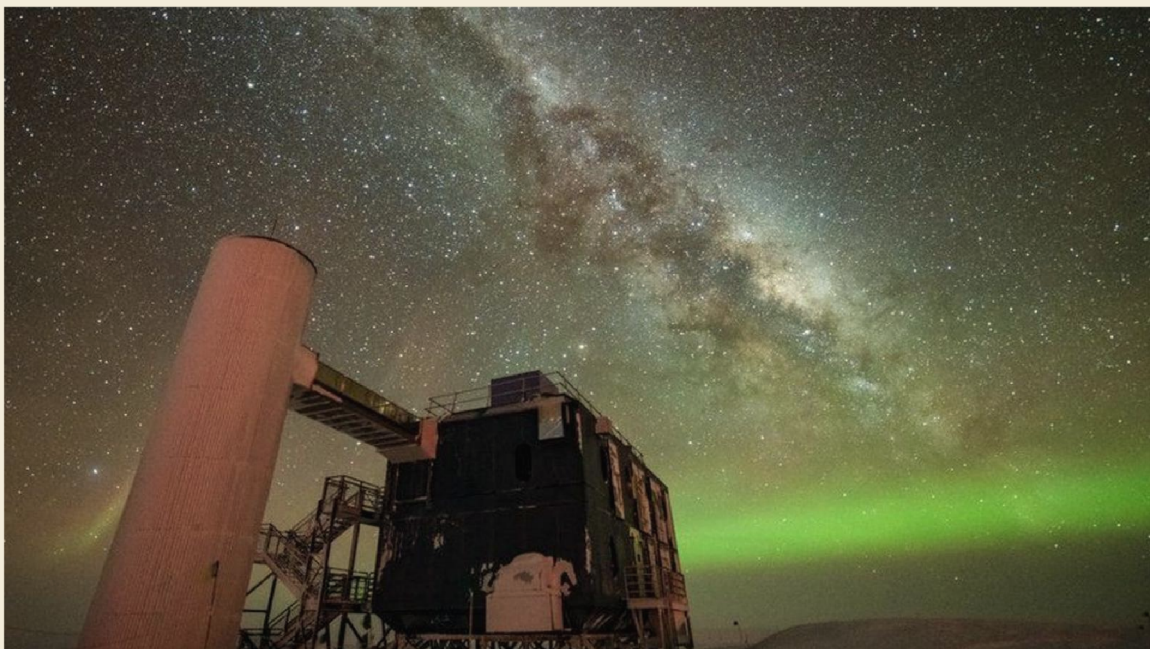

'Ghost particles' from space telescope wins physics Nobel

6 October 2026

Pallab Ghosh, Science correspondent



Yuya Makino/IceCube/NSF: Prof Francis Halzen won for “decisive contribution” to developing the IceCube Neutrino Observatory

Belgian-born physicist Prof Francis Halzen has won the Nobel Prize in Physics for his pioneering work on an observatory that detects particles from space, shedding light on the distant universe.

Halzen led the development of IceCube at the South Pole, which uses a cubic kilometre of Antarctic ice fitted with light sensors to detect particles called neutrinos. These carry information about violent, high-energy processes in the cosmos.

The Royal Swedish Academy of Sciences said “his vision and scientific leadership have been fundamental for the IceCube Neutrino Observatory”.

He first presented his vision for detecting neutrinos at the South Pole in 1988.

Halzen, who is now a US citizen at the University of Wisconsin–Madison, told a news conference that he was surprised his idea worked so well.

“I have to emphasise how lucky I was. Because when we started this project, everybody realised this was maybe a good idea, but very few thought it would work, including myself.”

Neutrinos are elusive subatomic particles. The Sun produces vast numbers of them: every second, billions pass harmlessly through an area the size of your fingernail.

But IceCube is searching for much higher-energy neutrinos, those produced by violent processes far beyond our Solar System — in distant galaxies.

It is composed of thousands of sensors on long cables that are drilled and frozen into a 1km cubic block of ice.

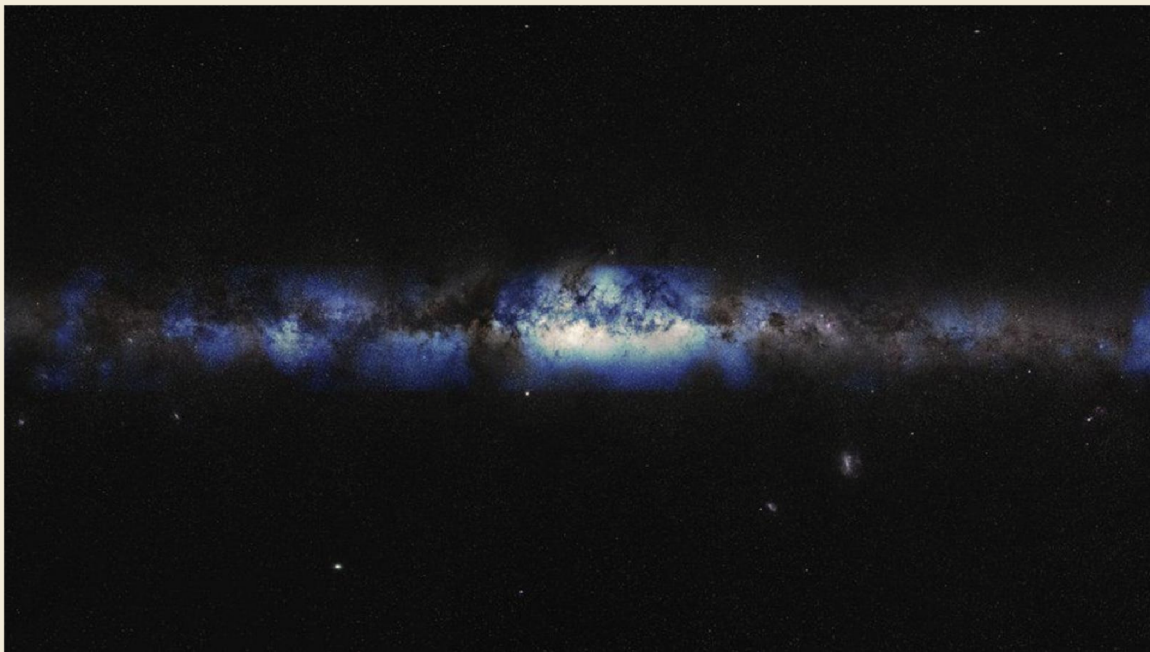
Neutrinos can be thought of as astronomical messengers that point to those fundamental events.

At the observatory, most neutrinos pass straight through the ice, but occasionally one interacts with it, producing particles that give off a telltale flash of light.

These rare signals carry clues to where the neutrinos came from and the extreme conditions that produced them — allowing scientists to investigate places that ordinary telescopes cannot easily see.

“Francis Halzen has led an international team of researchers and engineers who have provided us with a fantastic instrument,” said Prof Mark Pearce, Chair of the Nobel Committee for Physics.

“His tenacity and scientific vision has paved the way for a new kind of astronomy.”



IceCube/NSF: This blurry image is of the Milky Way, but it is composed of neutrino particles

On the rare occasions when a neutrino strikes an atomic nucleus, it produces fast-moving charged particles. It is these racing through the ice, which give off a blue glow, which sensitive light detectors can pick up.

The pattern and timing of that light allow scientists to estimate the direction the neutrino came from — helping them trace its origins in the cosmos. But these interactions are so rare that the observatory needed an enormous volume of clear ice.

Halzen's idea was to use a vast natural ice sheet in Antarctica. So began an audacious plan to build a neutrino observatory at the geographic South Pole, with thousands of light sensors embedded deep in the ice.

The neutrino's ghostly nature, the very thing that makes it so hard to detect, also makes it a powerful messenger from the distant Universe.

Neutrinos rarely interact with material in its way, so they can go right through dense regions that light cannot easily penetrate.

And because they have no electric charge, magnetic fields do not bend their paths through space. That means their direction of arrival can point scientists back towards their source.

Their energies carry further insight into the processes that created them - helping scientists investigate how exploding stars and the environments around giant black holes accelerate particles to extraordinary energies.

Louis Barson, director of science at the Institute of Physics in London, said the IceCube Observatory had enabled astronomers to investigate phenomena that conventional telescopes cannot see.

“The discoveries that have followed are helping us understand some of the most energetic and mysterious processes in the universe - and that’s thanks to Professor Halzen’s work,” he added.

Prof Subir Sarkar, physicist from the University of Oxford, quoted the French novelist Marcel Proust in describing his assessment of Halzen’s contribution to science:

“‘The only true voyage of discovery, the only fountain of Eternal Youth, would be not to visit strange lands but to possess other eyes, to behold the universe through the eyes of another, of a hundred others, to behold the hundred universes that each of them beholds, that each of them is’.

“Francis Halzen has led the IceCube collaboration on just such a voyage of discovery and opened a new window on to our Universe”.

生词标注（新闻高频 + 词汇第二义项）

词	词性	中文	备注
wins	v.	赢得、获得（奖项）	“wins physics Nobel” 获得诺贝尔物理学 奖；正文作 has won； 获奖、夺冠、胜选均 用 win，标题省略冠 词
pioneering	adj.	开创性的、先驱的	“his pioneering work on an observatory” 他在观 测站上的开创性工 作；动词 pioneer 开 辟、倡导
observatory	n.	天文台、观测站	“the IceCube Neutrino Observatory” 冰立方中 微子观测站；observe 观测
detect	v.	探测、检测到	“an observatory that detects particles from space” 探测来自太空 粒子的观测站；名词 detection 探测

词	词性	中文	备注
neutrino	n.	中微子	“particles called neutrinos” 被称为中微子的粒子；标题 ghost particles（幽灵粒子）即指它
elusive	adj.	难以捉摸的、难以捕捉的	“Neutrinos are elusive subatomic particles” 中微子是难以捕捉的亚原子粒子；动词 elude 躲开、使……无法得到
shed light on	phr.	揭示、使……得以了解	“shedding light on the distant universe” 揭示遥远宇宙的奥秘；新闻高频短语，≈ help explain
decisive	adj.	决定性的	“decisive contribution” 决定性贡献；动词 decide 决定，decision 决定（名词）
tenacity	n.	坚韧不拔、执着	“His tenacity and scientific vision” 他的坚韧与科学远见；形容词 tenacious
pave the way	phr.	为……铺平道路、奠定基础	“paved the way for a new kind of astronomy” 为一种新的天文学奠定基础
audacious	adj.	大胆的、敢于冒险的	“So began an audacious plan” 于是一项大胆的计划启动了；名词 audacity 大胆
embedded	v.	嵌入、深植	“sensors embedded deep in the ice” 深嵌冰层中的传感器；原形 embed
telltale	adj.	泄露踪迹的、揭示真相的	“a telltale flash of light” 一道泄露踪迹的闪光；telltale sign 明显的迹象

词	词性	中文	备注
ghostly	adj.	幽灵般的	“The neutrino’s ghostly nature” 中微子的“幽灵”特性；呼应标题 ghost particles
penetrate	v.	穿透、渗入	“regions that light cannot easily penetrate” 光难以轻易穿透的区域；名词 penetration 穿透
conventional	adj.	传统的、常规的	“conventional telescopes cannot see” 传统望远镜看不到的；反义 unconventional 非常规的
strikes	v.	击中、撞击	“when a neutrino strikes an atomic nucleus” 当中微子撞上原子核时；另义“罢工、袭击”
estimate	v.	估算、估计	“allow scientists to estimate the direction” 使科学家得以估算方向；名词 estimate 估算值（新闻高频第二义项）
give off	phr.	发出（光、热、气味等）	“give off a telltale flash of light” 发出泄露踪迹的闪光
carry	v.	携带、传递；刊登	“These carry information about violent, high-energy processes” 它们携带着关于剧烈高能过程的信息；报纸“刊登”报道也用 carry（第二义项）